

Persistence of human norovirus in reconstituted pesticides--pesticide application as a possible source of viruses in fresh produce chains.



The consumption of fresh produce is frequently associated with outbreaks of human norovirus (hNoV) disease. To prevent the contamination of fresh produce with hNoV, knowledge of the possible introduction sources of the viruses, such as water, is needed to be able to implement appropriate and efficient preventive measures. Contaminated water used to reconstitute pesticides could be a relevant source of infectious hNoV, determined by the initial level of virus contamination and the persistence of these viruses in reconstituted pesticides. We studied the persistence of hNoV GI.4, hNoV GII.4 and murine norovirus (MNV-1), the only culturable norovirus, in eight different pesticides after 0 and 2h. Virus concentrations were determined by reverse transcriptase PCR, and infectivity of MNV-1 was determined by endpoint dilutions followed by maximum likelihood estimations. MNV-1 was found to remain infectious in seven of the eight tested pesticides at the highest concentration applied in practice. In the presence of the insecticide Vertimec, MNV-1 infectivity decreased rapidly with a 1.9 log(10)-unit reduction at timepoint T(0). Also, the concentration of NoV GI.4 RNA decreased considerably with a 1.7 log(10)-unit reduction; whereas the detected PCR fragment of hNoV GII.4 remained stable. Assuming a similar persistence of infectious MNV-1 and hNoV we can conclude that water containing hNoV used to dilute pesticides may be an important source of infectious hNoV in fresh produce chains. The application of pesticides may therefore not only be a chemical hazard, but also a microbiological hazard for public health. The inclusion of antiviral substances in reconstituted pesticides may be appropriate to reduce the virological health risk posed by the application of pesticides. Copyright © 2012 Elsevier B.V. All rights reserved.